

# INSPIRE WP4 - Health economics

Analysis update

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may 2026

# Why?

- **we are here:** from call HORIZON-HLTH-2021-DISEASE-04-01, *“proposals under this topic should aim for delivering results that are directed, tailored towards and contributing to [... among other...] **reduce societal, healthcare and economic burden** associated with increasing demands of supportive, palliative, survivorship or end-of-life care services that is beneficial for citizens and preserves the sustainability of the health care systems.”*
- **I am here:**
  - **background:** MsC economics (in a past life), 1 post-graduate year Statistics, almost MsC Statistics (2 exams left)
  - project initially financed with Dr. Massimo Costantini (I didn't participate in first iterations/design); **support** for this part needed, since formally assigned to AUSL-RE

# HE analysis - the overall picture I

Data collection for economics analysis performed alongside a clinical trial is less “self-contained” if compared to statistical analysis. This below depicts the data flow

Main components (data) are three:

- 1 unit **costs**
- 2 **quantities**
- 3 **QOL/Utilities**

Three above constitutes the section of these slides too, where I'll give update discuss difficulties and decision taken (up to now).

Approach adopted so far:

- make it run, make it right, make it fast
- MVP (minimum viable product)

# Part I

## Cost

# Timeline I

**2025-06-24:** mail from Elena, unit cost collection request

| Home help / care worker / health care assistant                       |                         |             |   |
|-----------------------------------------------------------------------|-------------------------|-------------|---|
| A                                                                     | B                       | C           | D |
| Country: _____                                                        |                         |             |   |
| Institution: _____                                                    |                         |             |   |
| Currency to be considered for costs filled in below (eg: Euro): _____ |                         |             |   |
| Main/reference year for cost filled in below (eg: 2025): _____        |                         |             |   |
|                                                                       |                         |             |   |
|                                                                       |                         |             |   |
|                                                                       |                         |             |   |
| Service                                                               | Mean Comprehensive Cost | Metric/Unit |   |
| <b>1. Hospital services</b>                                           |                         |             |   |
| Hospital inpatient ward                                               |                         | per day     |   |
| Intensive/Critical care ward/unit                                     |                         | per day     |   |
| Specialist/rehabilitation ward/unit                                   |                         | per day     |   |
| <b>1. Other residential services</b>                                  |                         |             |   |
| Hospice/Palliative Care in-patient service                            |                         | per day     |   |
| Nursing or residential home                                           |                         | per day     |   |
| <b>2. Outpatient hospital services</b>                                |                         |             |   |
| Emergency ambulance service                                           |                         | per visit   |   |

# Timeline II

july-november 2025: unit cost collected from partners.  
january-february 2026: homogeneization

|                                                                                           |               |         |          |      |       |         |                     |                |             |                                                                                                       |                      |
|-------------------------------------------------------------------------------------------|---------------|---------|----------|------|-------|---------|---------------------|----------------|-------------|-------------------------------------------------------------------------------------------------------|----------------------|
| Institution: Charles university                                                           |               |         |          |      |       |         |                     |                |             |                                                                                                       |                      |
| Currency to be considered for costs filled in below: CZK                                  |               |         |          |      |       |         |                     |                |             |                                                                                                       |                      |
| Main/reference year for cost filled in below (eg: 2025)                                   |               |         |          |      |       |         |                     |                |             |                                                                                                       |                      |
|                                                                                           | Imported data |         |          |      |       |         | Provided by partner |                |             |                                                                                                       |                      |
| Service                                                                                   | value used    | imputed | currency | year | per   | source  | notes               | Mean cost      | Metric/Unit | Source                                                                                                | Note/Comments        |
| 1. Hospital services                                                                      |               |         |          |      |       |         |                     |                |             |                                                                                                       |                      |
| Hospital inpatient ward                                                                   | 561           | 0       | CZK      | 2025 | day   |         |                     | 561            | per day     | Czech ministry of health- <a href="https://mzd.gov.cz/v">https://mzd.gov.cz/v</a>                     |                      |
| Intensive/Critical care ward/unit                                                         | 4400          | 0       | CZK      | 2025 | day   |         |                     | 4400           | per day     | same                                                                                                  | there is a huge di   |
| Specialist/rehabilitation ward/unit                                                       | 2850          | 0       | CZK      | 2025 | day   |         |                     | 2850           | per day     | same                                                                                                  |                      |
| 2. Other residential services                                                             |               |         |          |      |       |         |                     |                |             |                                                                                                       |                      |
| Hospice/Palliative Care in-patient service                                                | 2400          | 0       | CZK      | 2025 | day   |         |                     | 2400           | per day     | same                                                                                                  |                      |
| Nursing or residential home                                                               | 1500          | 0       | CZK      | 2025 | day   |         |                     | 1500           | per day     | same                                                                                                  |                      |
| 3. Outpatient hospital services                                                           |               |         |          |      |       |         |                     |                |             |                                                                                                       |                      |
| Emergency ambulance service                                                               | 12000         | 1       | CZK      | 2025 | visit | FOR CH* |                     | 6000           | per visit   | <a href="https://www.zszhmp.cz/cer">https://www.zszhmp.cz/cer</a>                                     | fully covered for i  |
| Emergency department (A&E Dept., Accident and *                                           | 5500          | 1       | CZK      | 2025 | visit | FOR CH* |                     | 1000 - 10000   | per visit   | <a href="https://mzd.gov.cz/wp-cont">https://mzd.gov.cz/wp-cont</a>                                   | Depends on pres      |
| Exercise rehabilitation at the hospital (i.e. pulmona                                     | 560           | 0       | CZK      | 2025 | visit | FOR CH* |                     | 560            | per visit   | <a href="https://www.infoprolekare.cz">https://www.infoprolekare.cz</a>                               | This is the price c  |
| Specialist medical outpatient visit (e.g., cancer, specialist palliative, respiratory...) |               |         |          |      |       |         |                     |                | per visit   | <a href="https://mzd.gov.cz/wp-content/uploads/2024/">https://mzd.gov.cz/wp-content/uploads/2024/</a> |                      |
| Other non-medical hospital outpatient visit                                               |               |         |          |      |       |         |                     | x              | per visit   | x                                                                                                     |                      |
| Day hospital department (e.g., chemotherapy and *                                         | 500           | 0       | CZK      | 2025 | visit | FOR CH* |                     | Depends on tot | per visit   | <a href="https://mzd.gov.cz/wp-cont">https://mzd.gov.cz/wp-cont</a>                                   | If there is radiothe |
| 3. Community-based services                                                               |               |         |          |      |       |         |                     |                |             |                                                                                                       |                      |
| General Practitioner/Primary Care Physician                                               | 1000          | 0       | CZK      | 2025 | visit |         |                     | 1000           | per visit   | <a href="https://szv.mzcr.cz/vykon/">https://szv.mzcr.cz/vykon/</a>                                   | GPs are payed by     |

# Unit cost normalization: some general rules

## Standard year

Vast majority of unit costs refer to 2025 so this was chosen as *standard year*. Cost referred to other years (eg some from Italy and France) were moved to 2025 applying the project's long term inflation index (NICE's 3.5%).

## Multiple unit costs for same item

For a given unit cost if *different values* have been provided, we have to choose one (because we don't have the level of information in the CSRI to choose which one to adopt). As a general rule a simple *mean* was adopted at first instance, basically in order to get the "order of magnitude". The unique exception to this is *cost of drugs* where to get the long term unit cost we'll adopt the least amount applicable.

### 3) Community based services (unit of reference) I

In all countries unit cost for **community based services** (eg general practitioner, nurse etc) were provided on a **per visit** basis, as asked for. The only exception was Italy where data available were:

- **monthly** cost for GP;
- **hourly** cost for other professionals

*Per-visit* mean length (to obtain unit cost) was obtained by own estimates (hypothesizing and or asking colleagues):

- GP: hypothesizing 20h of visits per week, 15min each (as in betterb)
- 60 min visit: Physiotherapist, Occupational therapist, Speech & language therapist, Psychologist
- 30 min visit: General practice nurse, District/Public health/Community nurse, Dietitian, Psychiatrist, Rehabilitation physician/Physiatrist, Specialist palliative care doctor/consultant, Specialist palliative care nurse, Home palliative care / home hospice care service, Social worker, Home help / care worker / health care assistant



## 4) Informal care: wages I

Informal care hours will be valued using *mean wage* when adopting the society point of view (not health sector only) to estimate the cost associated to the program.

*Minimum wage* and *wage for unskilled worker* will not be considered due to low availability.

Mean-wages were imputed for *all countries* to have fully available and consistent estimates across all countries:

- *average full-time wage* is available from several sources
- for *average weekly worktime* we have both overall and by sex: for the moment I used mean male number of hours to have a better approximation of *full time work* number of hours (still slightly underestimate but better bindable to average full-time wage at previous point)

## 4) Informal care: wages II

| A          | B        | C           | D        | E                 | F                  | G                    | H                |
|------------|----------|-------------|----------|-------------------|--------------------|----------------------|------------------|
| country_id | currency | average_net | year_net | average_hours_all | average_hours_male | average_hours_female | mean_hourly_wage |
| cz         | CZK      | 40942       | 2025     | 34.5              | 36.5               | 32.1                 | 261.7            |
| dk         | DKK      | 34849       | 2024     | 28.9              | 31.4               | 26.2                 | 259.0            |
| fr         | EUR      | 2733        | 2024     | 30.9              | 33.4               | 28.2                 | 19.1             |
| it_re      | EUR      | 2017        | 2024     | 33.9              | 36.9               | 29.8                 | 12.8             |
| it_mi      | EUR      | 2017        | 2024     | 33.9              | 36.9               | 29.8                 | 12.8             |
| no         | NOK      | 41649       | 2024     | 26.6              | 28.7               | 24.3                 | 339.2            |
| uk_en      | GBP      | 2374        | 2025     | 31.2              | 34.5               | 27.6                 | 16.1             |
| uk_sc      | GBP      | 2374        | 2025     | 31.2              | 34.5               | 27.6                 | 16.1             |

## 5) Equipment: purchase vs rent I

In some equipment purchase (not rental) cost was available. This was turned to “equivalent” monthly rent by applying a standard 5-year usage and actualizing 60 common expenditures, one per month, to make them equivalent (after actualization) to purchase cost.

$$\text{Purchase cost} = \frac{\text{MR}}{(1+i)^{1/12}} + \frac{\text{MR}}{(1+i)^{2/12}} + \dots + \frac{\text{MR}}{(1+i)^{60/12}}$$

With  $i = 0.035$  yearly interest rate of project (3.5% as above) and given purchase cost only unknown is monthly equivalent rent (MR) which can be found by

$$\text{MR} = \frac{\text{Purchase cost}}{\left[ \frac{1}{(1+i)^{1/12}} + \frac{1}{(1+i)^{2/12}} + \dots + \frac{1}{(1+i)^{60/12}} \right]}$$

## 5) Equipment: purchase vs rent II

| <u>cost_id</u> | <u>cost_description</u>  | <u>country_id</u> | <u>purchase_cost</u> | <u>currency</u> | <u>year</u> | <u>monthly_rent</u> |
|----------------|--------------------------|-------------------|----------------------|-----------------|-------------|---------------------|
| 505            | Walking stick            | fr                | 12.7                 | EUR             | 2025        | 0.2                 |
| 501            | Ambulator oxygen         | no                | 140.1                | NOK             | 2025        | 2.5                 |
| 502            | Long term oxygen therapy | no                | 4102.0               | NOK             | 2025        | 74.5                |
| 503            | Non-invasive ventilation | no                | 3208.0               | NOK             | 2025        | 58.3                |
| 508            | Feeding pump             | no                | 3500.0               | NOK             | 2025        | 63.6                |

# Other rules I

## Other special rules

`it_mi` inherited from `it_re`, `uk_sc` inherited `uk_en` (only where needed)

## Currency and purchase power conversion

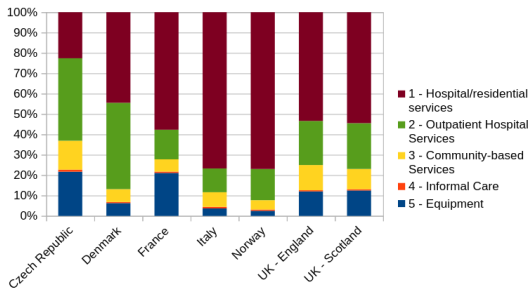
To normalize for different costs of living different quantities (all in 2025 EUR) purchasing power parity was applied, with the same methodology used for BETTER-B or Diadic.

# Final costs (EUR 2025)

| description                  | <u><a href="#">cz</a></u> | dk           | fr            | <u><a href="#">it_mi</a></u> | <u><a href="#">it_re</a></u> | no           | <u><a href="#">uk_en</a></u> | <u><a href="#">uk_sc</a></u> |
|------------------------------|---------------------------|--------------|---------------|------------------------------|------------------------------|--------------|------------------------------|------------------------------|
| Hospital inpatient ward      | 27.7                      | 819.3        | <b>1541.6</b> | <b>683.9</b>                 | 683.9                        | 1110.3       | 659.4                        | <b>659.4</b>                 |
| Intensive/Critical care ward | 217.3                     | 2560.4       | 938.0         | <b>2082.9</b>                | 2082.9                       | 5759.9       | 2109.7                       | <b>2109.7</b>                |
| Specialist/rehab ward/unit   | 140.8                     | 819.3        | <b>592.3</b>  | <b>650.3</b>                 | 650.3                        | 1596.1       | 620.9                        | <b>620.9</b>                 |
| Hospice                      | 118.5                     | <b>372.1</b> | 451.0         | <b>248.9</b>                 | 248.9                        | 1318.5       | 505.4                        | 399.2                        |
| Nursing or residential home  | 74.1                      | <b>164.0</b> | 62.4          | <b>135.0</b>                 | 135.0                        | <b>203.3</b> | 174.7                        | <b>215.1</b>                 |
| Emergency ambulance          | <b>592.6</b>              | <b>189.1</b> | <b>77.5</b>   | <b>87.0</b>                  | 87.0                         | <b>234.4</b> | 288.4                        | <b>288.4</b>                 |
| Emergency dept (A&E)         | <b>271.6</b>              | 1194.2       | 349.1         | <b>161.4</b>                 | 161.4                        | 329.6        | 124.9                        | <b>124.9</b>                 |
| Hospital rehab               | 27.7                      | <b>56.0</b>  | 98.6          | <b>13.4</b>                  | 13.4                         | 69.4         | <b>60.1</b>                  | <b>60.1</b>                  |
| Specialist                   | <b>64.7</b>               | 876.0        | 25.2          | <b>24.9</b>                  | 24.9                         | 208.2        | 206.7                        | <b>206.7</b>                 |
| Other outpatient visit       | <b>53.0</b>               | <b>137.5</b> | <b>97.4</b>   | <b>24.9</b>                  | 24.9                         | 208.2        | 147.5                        | <b>147.5</b>                 |
| Day hospital                 | 24.7                      | 2063.0       | <b>254.7</b>  | <b>265.1</b>                 | <b>265.1</b>                 | 936.9        | 823.8                        | <b>823.8</b>                 |
| General Practitioner         | 49.4                      | 17.3         | 28.5          | <b>32.1</b>                  | <b>32.1</b>                  | <b>49.9</b>  | 43.2                         | <b>43.2</b>                  |
| General practice nurse       | <b>11.0</b>               | 17.3         | <b>20.3</b>   | <b>23.5</b>                  | <b>23.5</b>                  | <b>35.5</b>  | 13.2                         | 33.7                         |
| District/community nurse     | <b>11.0</b>               | <b>28.6</b>  | <b>20.3</b>   | <b>23.5</b>                  | <b>23.5</b>                  | <b>35.5</b>  | 20.7                         | 33.7                         |
| Physiotherapist              | 27.7                      | 53.1         | <b>19.5</b>   | <b>13.5</b>                  | <b>13.5</b>                  | <b>34.1</b>  | 25.4                         | 33.7                         |
| Occupational therapist       | 27.7                      | <b>23.6</b>  | <b>16.7</b>   | <b>13.0</b>                  | <b>13.0</b>                  | <b>29.3</b>  | 25.4                         | 33.7                         |
| Speech & language therapist  | 9.9                       | <b>23.6</b>  | <b>16.7</b>   | <b>13.5</b>                  | <b>13.5</b>                  | <b>29.3</b>  | 25.4                         | 33.7                         |

# Final costs (sum by section, EUR 2025)

| Category                          | Czech Republic | Denmark | France | Italy | Norway | UK - England | UK - Scotland |
|-----------------------------------|----------------|---------|--------|-------|--------|--------------|---------------|
| 1 - Hospital/residential services | 578            | 4735    | 3585   | 3801  | 9988   | 4070         | 4004          |
| 2 - Outpatient Hospital Services  | 1034           | 4516    | 902    | 577   | 1987   | 1651         | 1651          |
| 3 - Community-based Services      | 364            | 671     | 374    | 355   | 597    | 944          | 735           |
| 4 - Informal Care                 | 25             | 59      | 40     | 40    | 79     | 43           | 43            |
| 5 - Equipment                     | 555            | 669     | 1311   | 181   | 325    | 918          | 918           |



Analisi missingess



## Part II

### QOL and utilities

So far we've talked only about unit cost and quantities ...

Cost-utility analysis (primary)

$$ICER = \frac{\text{diff in QALY between arms}}{\text{diff in (unit cost} \cdot \text{quantities) btw arms}}$$

Cost-effectiveness analysis (secondary)

$$ICER = \frac{\text{diff in FACT-G between arms}}{\text{diff in (unit cost} \cdot \text{quantities) btw arms}}$$

Differences in numerator ...

# Utility basics

- definition
  - in *general*: measure of a certain person's **satisfaction** from a certain state of the world
  - in *our context*: *self evaluation* of satisfaction regarding health condition
- ranges and interpretation (again, in our context):

$$u_t = \begin{cases} 1 & \text{perfect health} \\ 0 & \text{equivalent to death} \\ < 0 & \text{worst than being dead} \end{cases} \quad \text{at time } t$$

- sources: obtained from QOL questionnaires
- mappings:
  - EQ-5D-5L: direct mapping
  - FACT-G: indirect mapping applying Meregaglia et al. 2019

**TODO:** inserire le equazioni della meregaglia?

| Timepoint | NA  | mean utility |
|-----------|-----|--------------|
| Baseline  | 0   | 0.431522     |
| W4        | 66  | 0.437461     |
| W8        | 101 | 0.442907     |
| W16       | 169 | 0.442532     |

Table: Mean raw utility, as of january 2026

# From utility to QALY

QALY = quality adjusted life years

Assume you have the measurement of utility in time: QALY is the area under the (function of) utility in time.

Some examples with constant utility:

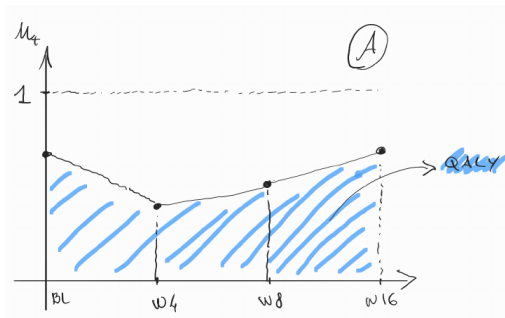
- a person who evaluates him/herself as *perfectly healthy* (utility=1) and is observed for 1 year, at the end of the year will have a  $QALY = 1$  (area of  $utility \cdot year = 1 \cdot 1 = 1$ )
- a person who has utility of 0.5, observed for 2 years will have a  $QALY = 1$  at the end

Thus the theoretical maximum QALY a patient can have in our study is

$$1 \cdot \frac{16}{52} = 0.31$$

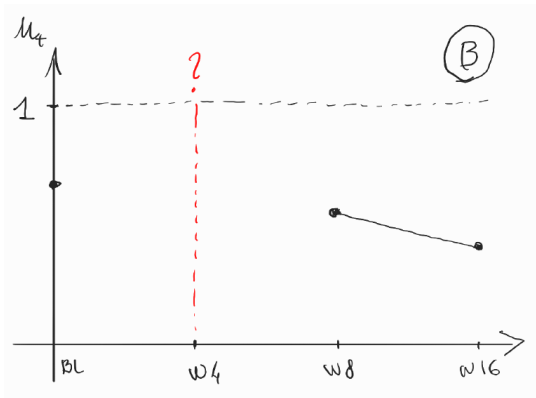
# QALY I

When utility is not constant in time (as often is) we can interpolate available observation/points. Eg wrt full observation available



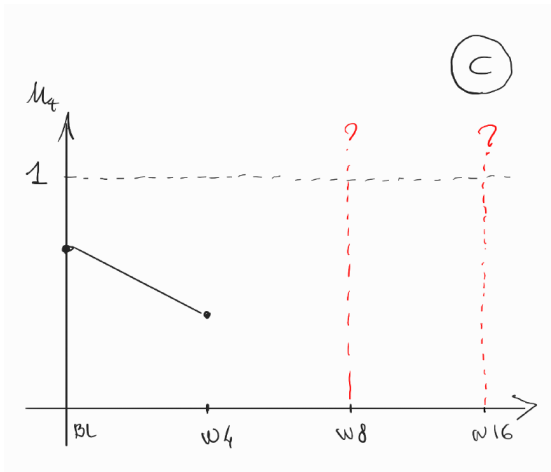
# QALY II

No show during fup



# QALY III

No show at the end of follow up





Frequencies (as of january)

# Next steps I

We have to:

- 1 prepare queries (up to january not that much)
- 2 complete the analysis;
- 3 write paper following Husereau et al. 2022;

## Part III

### Misc

# References I



Husereau, Don et al. (Jan. 2022). "Consolidated Health Economic Evaluation Reporting Standards (CHEERS) 2022 Explanation and Elaboration: A Report of the ISPOR CHEERS II Good Practices Task Force". en. In: *Value in Health* 25.1, pp. 10–31. ISSN: 10983015. DOI: 10.1016/j.jval.2021.10.008. URL: <https://linkinghub.elsevier.com/retrieve/pii/S1098301521017952> (visited on 07/31/2025).



Meregaglia, Michela et al. (Mar. 2019). "Mapping Health-Related Quality of Life Scores from FACT-G, FAACT, and FACIT-F onto Preference-Based EQ-5D-5L Utilities in Non-Small Cell Lung Cancer Cachexia". In: *The European Journal of Health Economics* 20.2, pp. 181–193. ISSN: 1618-7598, 1618-7601. DOI: 10.1007/s10198-017-0930-6. (Visited on 05/07/2026).